Spring 2021 Real Analysis II — Homework 5

FirstName LastName
ID: 123-45-6789

1. Let X and Y be Banach spaces and $T \in L(X, Y)$. Suppose T is surjective, and there exists $m > 0$ such that $\|Tx\| \geq m\|x\|$ for all $x \in X$. Show that T^{-1} exists, $T^{-1} \in L(Y, X)$, and $\|T^{-1}\| \leq 1/m$.

2. Let X be a Hilbert space and $T \in L(X)$. Suppose there exists $m > 0$, such that $|\langle Tx, x\rangle| \geq m\|x\|^2$ for all $x \in X$. Show that T^{-1} exists and $T^{-1} \in L(X)$.

3. Let X and Y be B* spaces and D a linear subspace of X. Suppose $T : D \to T(D) \subset Y$ is linear. Prove that the following statements hold:

   (a) If T is continuous and D is closed, then T is a closed operator.

   (b) Suppose T is continuous and closed. Then D is closed provided that Y is complete.

   (c) If T is closed and injective, then T^{-1} is closed.

   (d) If X is complete, T is closed and injective, $T(D)$ is dense in Y, and T^{-1} is continuous, then $T(D) = Y$.

4. Use Corollary 3.35 to show that $(C([0, 1]), \|\cdot\|_1)$ is not a Banach space, where $\|f\|_1 := \int_0^1 |f(t)|\,\mathrm{d}t$.

5. Let $(X, \|\cdot\|)$ be a Banach space and $p : X \to \mathbb{R}$ satisfy

   - (Nonnegative) $p(x) \geq 0$ for all $x \in X$.
   - (Positive homogeneous) $p(\lambda x) = \lambda p(x)$ for all $\lambda > 0$ and $x \in X$.
   - (Triangle inequality) $p(x_1 + x_2) \leq p(x_1) + p(x_2)$ for all $x_1, x_2 \in X$.
   - (Lower semicontinuous) If $x_k \to x$, then $\liminf_{k\to\infty} p(x_k) \geq p(x)$.

   Show that there exists $M > 0$ such that $p(x) \leq M\|x\|$ for all $x \in X$.

   [Hint: define $\|x\|_1 := \|x\| + \max(p(x), p(-x))$, show that $(X, \|\cdot\|_1)$ is a Banach space, and then apply Corollary 3.35 to show that $\|\cdot\|_1$ and $\|\cdot\|$ are equivalent.

6. Let $l^1 := \{x = (\xi_1, \xi_2, \dots) : \sum_{k=1}^{\infty} |\xi_k| < \infty,\ \xi_k \in \mathbb{R}\}$ and $a = (a_1, a_2, \dots)$ a real sequence. Suppose $\sum_{k=1}^{\infty} a_k\xi_k$ converges for any $x \in l^1$. Show that $f : l^1 \to \mathbb{R}$ defined by $f(x) := \sum_{k=1}^{\infty} a_k\xi_k$ is a bounded linear functional, and $\|f\| = \sup_{k\in\mathbb{N}} |a_k|$.

7. Let X and Y be Banach spaces and D a linear subspace of X. Let $T : D \to T(D) \subset Y$ be a linear closed operator. Denote $Z := \{x \in D : Tx = 0\}$. Show that the following statements hold.

   (a) Z is a closed linear subspace of X.

   (b) $Z = \{0\}$ and $T(D)$ is closed in Y if and only if there exists $a > 0$ such that $\|x\| \leq a\|Tx\|$ for all $x \in D$.

8. Let X and Y be Banach spaces and $T \in L(X,Y)$. Denote $Z := \{x \in X : Tx = 0\}$, $d(x,Z) := \inf_{z\in Z} \|x - z\|$, and
$$\gamma(T) := \inf_{x\notin Z} \frac{\|Tx\|}{d(x,Z)}.$$
Show that the following statements hold:

   (a) $\gamma(T) \leq \|T\|$.

   (b) If T^{-1} exists, then $\gamma(T) = \frac{1}{\|T^{-1}\|}$.

9. Let X be a Hilbert space and $a : X \times X \to \mathbb{R}$ a bilinear functional that satisfies

   (a) There exists $M > 0$ such that $|a(x,y)| \leq M\|x\|\|y\|$ for any $x, y \in X$.

   (b) There exists $\delta > 0$ such that $|a(x,x)| \geq \delta\|x\|^2$ for any $x \in X$.

   Show that for any $f \in X^*$, there exists a unique $y_f \in X$ such that $a(x, y_f) = f(x)$ for any $x \in X$. Moreover y_f continuously depends on f.

10. Let $\Omega \subset \mathbb{R}^2$ be an open bounded set with smooth boundary and $\alpha : \Omega \to \mathbb{R}$ a bounded measurable function satisfying $\alpha_0 \leq \alpha$ for some constant $\alpha_0 > 0$. Suppose $f \in L^2(\Omega)$ and denote
$$a(u,v) := \int_\Omega (\nabla u \cdot \nabla v + \alpha uv)\,\mathrm{d}x\,\mathrm{d}y \qquad \text{and} \qquad F(v) := \int_\Omega fv\,\mathrm{d}x\,\mathrm{d}y$$
for any $u, v \in H^1(\Omega)$. Show that there exists a unique $u \in H^1(\Omega)$ such that $a(u,v) = F(v)$ for any $v \in H^1(\Omega)$. [Hint: recall that $(H^1(\Omega), \|\cdot\|_1)$ where $\|u\|_1 := (\int_\Omega |u|^2 + |\nabla u|^2\,\mathrm{d}x\,\mathrm{d}y)^{1/2}$ is a Hilbert space.]